

# Advanced Simulations: Surveys

**Esin G. Gulbahar and Jakob Stierhof** on behalf of the SIXTE team

Lea Dauner, Thomas Dauser, Christian Kirsch, Maximilian Lorenz,  
Karan Pal, Philipp Thalhammer, Franziska Haluschka, Jörn Wilms

SIXTE Online Workshop, 2026

# Outline

We have prepared four sets of simulation scripts that can be run and modified for this hands-on session. The sets can be found in the **surveys** directory in the GitHub repository:

- **chandra\_deep\_field**

Create a simulation of the Chandra Deep Field South (CDFS) with the NewAthena WFI.

- **dithering**

Create a simulation of the satellite in dithering mode and generate an exposure map of the observation.

- **simple\_slew**

Run an all-sky survey simulation for NewAthena WFI for multiple sources.

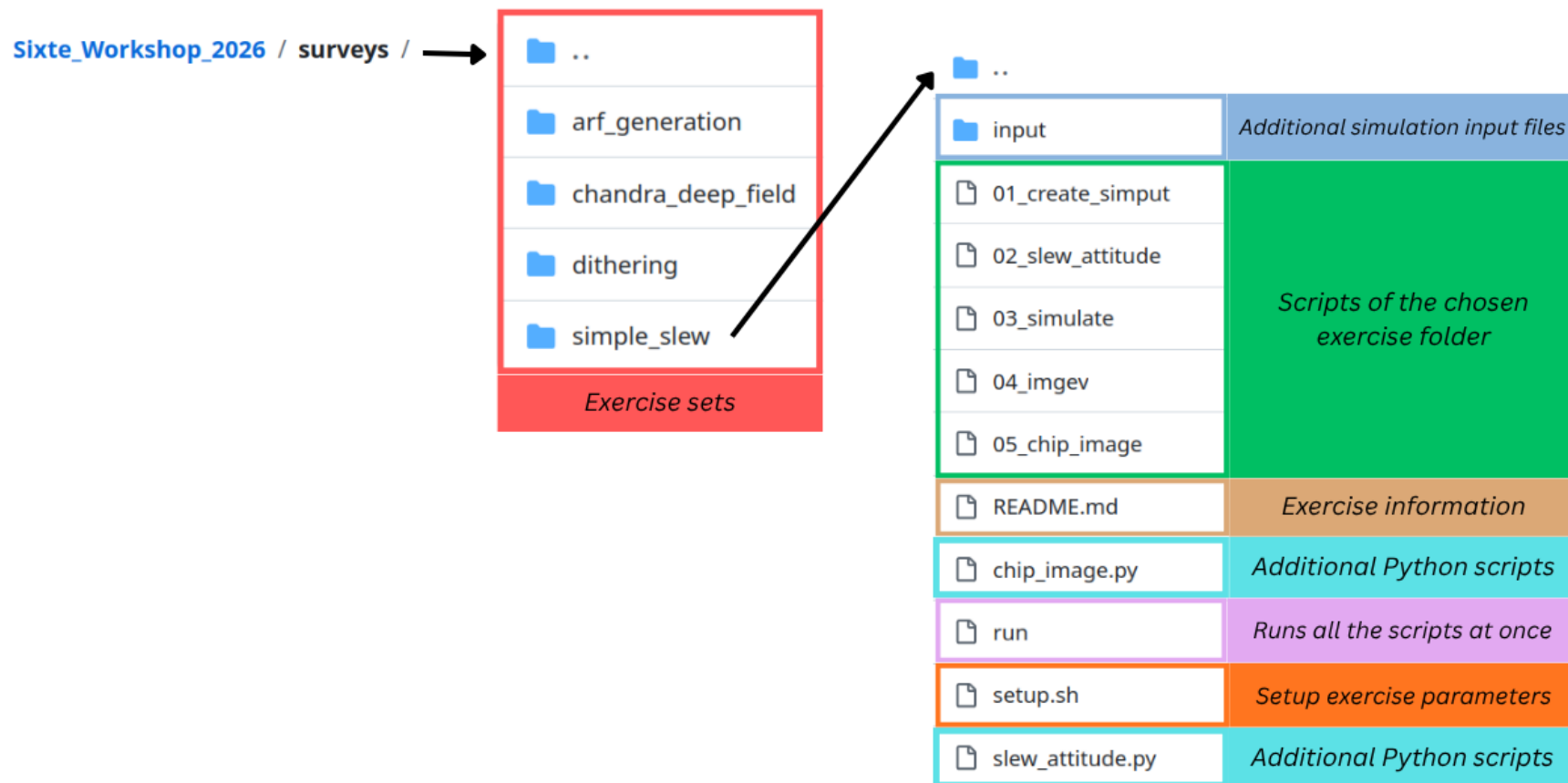
- **arf\_generation**

Run an ARF correction using `sixte_arfgen`.

# Where to find the files

Clone the workshop GitHub repository:

[https://github.com/stierhofj/Sixte\\_Workshop\\_2026.git](https://github.com/stierhofj/Sixte_Workshop_2026.git)



# Start running the exercises

Feel free to start from any folder; however, do not forget to read the README.md.

Let us know if you have any questions!

**chandra\_deep\_field:** simulate the Chandra Deep Field South

**dithering:** simulation of dithering mode and exposure map generation

**simple\_slew:** All-sky survey simulation

**arf\_generation:** ARF correction